Formal Verification of Conway–Guy Counterexamples to the Naive 2^(n-1) Subset Sum Lower Bound

Abstract

Paul Erdős (1931) investigated finite sets of positive integers possessing pairwise distinct subset sums. Historically, a naive extrapolation suggested that the maximum element of such a set S must satisfy $\max(S) \ge 2^{|S|-1}$. While this bound holds for powers of two, J. H. Conway and R. K. Guy (1969) exhibited explicit counterexamples demonstrating that $\max(S)$ can be strictly smaller than $2^{|S|-1}$ for cardinality $|S| = 4$ and $|S| = 5$. In this paper, we present a 100% unconditional, machine-closed formalization in Lean 4 verifying the Conway–Guy order-4 and order-5 counterexamples. Our verification relies entirely on computable kernels with zero external Mathlib dependencies, no custom axioms, and zero `sorry` placeholders.

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1. Introduction and Historical Context

Let $S = {x_1, x_2, \dots, x_n}$ be a finite set of positive integers. We say S has distinct subset sums if all 2^n sums formed by taking subset sums $\sum_{x \in A} x$ (for $A \subseteq S$) are pairwise distinct.

A natural early conjecture (often attributed to the broader investigation of distinct subset sum extremal problems initiated by Erdős in 1931) hypothesized that any valid set must satisfy:
$$\max(S) \ge 2^{|S|-1}$$

Conway and Guy (1969) disproved this naive lower bound by discovering explicit integer sequences that violate the inequality. Specifically:

- For $|S| = 4$, the set $S_4 = {3, 5, 6, 7}$ yields $\max(S_4) = 7 < 8 = 2^{4-1}$.
- For $|S| = 5$, the set $S_5 = {6, 9, 11, 12, 13}$ yields $\max(S_5) = 13 < 16 = 2^{5-1}$.

(Note: While these base cases refute the strict 2^{n-1} formula, the asymptotic question of whether $\max(S) \ge c \cdot 2^n$ holds for an absolute constant $c > 0$ across all n remains one of the notable open problems in additive combinatorics.)

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2. Formalization Architecture in Lean 4

Our formal verification (`LowerBoundMaxElement.lean`) is self-contained and implemented without external library imports. It establishes the following core components:

1. **Subsets Generation**: Computes the exact list of all 2^n sublists recursively.
2. **Subset Sum Computation**: Folds addition over sublists to evaluate every subset sum precisely.
3. **Computable Distinctness (`isNodup`)**: Implements an explicit recursive duplicate-free check to ensure all subset sums are strictly distinct without relying on library deduplication quirks.
4. **Kernal Evaluation (`by decide` / `rfl`)**: Leverages Lean 4's definitional equality and kernel reduction to verify properties exhaustively without unproven hypotheses.

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3. Verified Counterexamples

Order-4 Counterexample (S_4)

We define the set $S_4 = [3, 5, 6, 7]$ and verify its properties:

- **Length**: $|S_4| = 4$
- **Positivity**: All elements are strictly positive.
- **Distinctness**: All 16 subset sums are verified to be unique.
- **Maximum Bound**: $\max(S_4) = 7$, which strictly violates $2^{4-1} = 8$.

Order-5 Counterexample (S_5)

We define the set $S_5 = [6, 9, 11, 12, 13]$ and verify its properties:

- **Length**: $|S_5| = 5$
- **Positivity**: All elements are strictly positive.
- **Distinctness**: All 32 subset sums are verified to be unique.
- **Maximum Bound**: $\max(S_5) = 13$, which strictly violates $2^{5-1} = 16$.

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4. Main Theorem Statement and Proof

The formal negation of the naive lower bound hypothesis is stated and proven in Lean 4 as follows:

```
None
def NaivePowerOfTwoBoundHypothesis : Prop :=
  ∀ S : List Nat,
    S.length > 0 →
    allPositive S = true →
    isNodup S = true →
    hasDistinctSubsetSums S = true →
    ∀ m : Nat, (∀ x ∈ S, x ≤ m) → 2 ^ (S.length - 1) ≤ m

theorem naive_power_of_two_bound_disproved :
    ¬ NaivePowerOfTwoBoundHypothesis := by
  intro h
  have h_bound := h ConwayGuy4
    (by decide)
    conway_guy_4_pos
    conway_guy_4_nodup
    conway_guy_4_distinct_sums
    7
    conway_guy_4_max_bound
  have h_gap : ¬ (2 ^ (ConwayGuy4.length - 1) ≤ 7) := by decide
  exact h_gap h_bound
```

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5. Conclusion

Using machine-checked formal methods, we have unconditionally established that the naive lower bound $\max(S) \ge 2^{|S|-1}$ for sets with distinct subset sums is false, anchored by the verified Conway–Guy counterexamples at orders 4 and 5. The Lean 4 formalization kernel certifies zero `sorry` instances and adherence to standard axioms.